

Vieta's formulas

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If you ain't first, you're last!

Ricky Bobby. He was presumably talking about two-player games.

1 Theory

In this lecture, we introduce Vieta's formulas. Throughout, assume *roots* means *roots with multiplicity*. For example, the sum of the roots of the polynomial $(x - 5)^2(x + 4)$ is $5 + 5 - 4 = 6$.

Theorem 1.1 (Fundamental theorem of algebra). *For every polynomial P with degree n and complex coefficients, there exist unique complex a and $r_1, r_2, \dots, r_n$ such that*

$$P(x) = a \prod_{k=1}^n (x - r_k)$$

for all $x \in \mathbb{C}$. Basically, you can completely decompose polynomials over the complex numbers.

Theorem 1.2 (Vieta's formulas, quadratic case). *Let $P(x) = ax^2 + bx + c$ have roots r_1 and r_2 . Then $r_1 + r_2 = -\frac{b}{a}$ and $r_1 r_2 = \frac{c}{a}$.*

I'm now going to explain the general form of Vieta's formulas with examples, since the statement is kind of disgusting.

The "stereotypical" Vieta question will give you a polynomial, and ask you to find some symmetric expression in terms of the roots of that polynomial. For these, it's usually a good idea to write down the symmetric polynomials in terms of the roots and their values, then use algebraic manipulation to find whatever you need to.

Example 1.3. Let a, b , and c be the roots of $2x^3 - 4x^2 - 6x + 1$. Find

- $a + b + c$,
- $ab + bc + ca$,
- abc ,
- $a^2 + b^2 + c^2$,
- $a^2b + a^2c + b^2c + b^2a + c^2a + c^2b$,
- $a^3 + b^3 + c^3$,
- and $a^2b^2 + b^2c^2 + c^2a^2$.

2 Exercises

Exercise 2.1 (Putnam 1977). Consider all lines that meet the graph of $y = 2x^4 + 7x^3 + 3x^2 - 5$ in four distinct points, (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x_4, y_4) . Show that

$$\frac{x_1 + x_2 + x_3 + x_4}{4}$$

is independent of the line and find its value.

Exercise 2.2. Find the sum of the reciprocals of the roots of the polynomial $2x^3 + 2x + 3$.

Exercise 2.3 (BMT General 2021/12). Let a , b , and c be the solutions of the equation $x^3 - 3 \cdot 2021^2 x = 2 \cdot 2021^3$. Compute $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

Exercise 2.4 (BMT General 2021/14). Let $r_1, r_2, \dots, r_{47}$ be the roots of $x^{47} - 1 = 0$. Compute

$$\sum_{i=1}^{47} r_i^{2020}.$$

Exercise 2.5 (2002 AMC 12A/12). Both roots of the quadratic equation $x^2 - 63x + k = 0$ are prime numbers. Find all possible values of k .

Exercise 2.6 (2000 AMC 10/24). Let f be a function for which $f(\frac{x}{3}) = x^2 + x + 1$. Find the sum of all values of z for which $f(3z) = 7$.

Exercise 2.7 (2002 AMC 12B/6). Suppose that a and b are nonzero real numbers, and that the equation $x^2 + ax + b = 0$ has distinct solutions a and b . What is the ordered pair (a, b) ?

Exercise 2.8 (2006 AMC 10B/14). Let a and b be the roots of the equation $x^2 - mx + 2 = 0$. Suppose that $a + \frac{1}{b}$ and $b + \frac{1}{a}$ are the roots of the equation $x^2 - px + q = 0$. What is q ?

Exercise 2.9. Let a , b , c , x , and y be five real numbers such that $a^3 + ax + y = 0$, $b^3 + bx + y = 0$ and $c^3 + cx + y = 0$. If a , b , and c are all distinct, compute $a + b + c$.

Exercise 2.10. Find the sum of the fifth powers of the roots of $x^5 - 20x + 23$.

Exercise 2.11 (CHMMC). Suppose $f(x) = x^3 + x - 1$ has roots α , β , and γ . What is

$$\frac{\alpha^3}{1-\alpha} + \frac{\beta^3}{1-\beta} + \frac{\gamma^3}{1-\gamma}?$$

Exercise 2.12 (2003 AIME II/9). (★) Consider the polynomials $P(x) = x^6 - x^5 - x^3 - x^2 - x$ and $Q(x) = x^4 - x^3 - x^2 - 1$. Given that z_1 , z_2 , z_3 , and z_4 are the roots of Q , compute $P(z_1) + P(z_2) + P(z_3) + P(z_4)$.

Exercise 2.13 (2008 AIME II/7). (★) Let r , s , and t be the three roots of the equation

$$8x^3 + 1001x + 2008 = 0.$$

Find $(r+s)^3 + (s+t)^3 + (t+r)^3$.

Exercise 2.14 (SMT Algebra 2019/5). (★) Let $f(x) = 36x^4 - 36x^3 - x^2 + 9x - 2$. Then let the four roots of $f(x)$ be r_1 , r_2 , r_3 , and r_4 . Find the value of

$$(r_1 + r_2 + r_3)(r_1 + r_2 + r_4)(r_1 + r_3 + r_4)(r_2 + r_3 + r_4).$$

Exercise 2.15 (HMMT November Guts 2019/18). (★) The polynomial $x^3 - 3x^2 + 1$ has three real roots r_1 , r_2 , and r_3 . Compute

$$\sqrt[3]{3r_1 - 2} + \sqrt[3]{3r_2 - 2} + \sqrt[3]{3r_3 - 2}.$$

Exercise 2.16. (★) Let x_1 , x_2 , and x_3 be the three roots of the polynomial $x^3 + 3x + 1$. Compute

$$\frac{x_1^2}{(5x_2 + 1)(5x_3 + 1)} + \frac{x_2^2}{(5x_3 + 1)(5x_1 + 1)} + \frac{x_3^2}{(5x_1 + 1)(5x_2 + 1)}.$$

Exercise 2.17 (CMIMC ANT 2019/6). (★★) Let a , b , and c be the distinct solutions to the equation $x^3 - 2x^2 + 3x - 4 = 0$. Find the value of

$$\frac{1}{a(b^2 + c^2 - a^2)} + \frac{1}{b(c^2 + a^2 - b^2)} + \frac{1}{c(a^2 + b^2 - c^2)}.$$

Exercise 2.18 (HMMT Guts 2023/31 extraction). (★★) Let the roots of $x^3 - x - 1$ be a , b , and c . Compute

$$\left((a - b)(b - c)(c - a)\right)^2.$$